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Systematic dissection of endogenous A-to-I RNA editing reveals substrate- and target-level determinants of predictability

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Endogenous adenosine-to-inosine RNA editing reflects at least three scales: whether an RNA duplex is intrinsically competent, whether that substrate is realized and encountered in a cell, and which adenosine within it is selected. The information that resolves these scales in predictive models is unknown. We systematically evaluated biological annotations, supervision, representations, and integration strategies on 884 human inverted-Alu substrates while quarantining the published pair-disjoint test. Editing burden was predictable at the substrate level, and inter-arm geometry added reproducible information beyond graph-derived summaries. Yet tested bulk RBP, transcript, repeat, and chromatin measurements did not estimate this realized state precisely enough to improve the leading site classifier. Two principles did. Sampling substrates before sites prevented site-rich duplexes from dominating learning, independently of the benefit from recovering additional high- and low-editing observations. A target- and helix-partner-aligned representation recovered site-level selectivity diluted by whole-duplex pooling; an identical encoder shifted 96 nucleotides away failed. Independently trained global and local views were complementary, whereas a preregistered end-to-end integration did not surpass their fixed equal-logit combination. Frozen before test access, the resulting ADARPro predictor improved macro AUROC, AUPRC, and F1 by 0.0277, 0.0261, and 0.0253 over reproduced published AdarEdit in a single canonical evaluation, with positive pair-bootstrap intervals in every endpoint and fewer false positives at matched recall. Endogenous editing predictability is therefore hierarchical: substrate realization is detectable but incompletely measured by current bulk proxies, whereas substrate-aware learning and target-centered representation supply actionable information.

A-to-I RNA editing | ADAR | inverted Alu | RNA structure | machine learning

Significance

Why one adenosine is edited while another apparently suitable site is not remains unclear. We separated this problem into the competence and realization of a double-stranded RNA substrate and the selection of a nucleotide within it. Editing burden at the substrate level was measurable, but current tissue-average molecular annotations did not resolve that state well enough to improve site prediction. In contrast, learning equally from RNA substrates and explicitly representing each target with its paired nucleotide provided reproducible information. A predictor built from these principles improved every held-out tissue endpoint. The results define what current data can explain and identify the matched cellular measurements needed next.

Adenosine deaminases acting on RNA (ADARs) convert adenosine to inosine in double-stranded RNA (dsRNA), diversifying transcripts while preventing endogenous duplexes from activating antiviral pathways (1–5). Most detected human editing occurs in paired Alu elements, whose abundance and orientation create imperfect intramolecular duplexes (6–10). Nevertheless, editing is selective: nearby adenosines within the same predicted structure can differ sharply, and the same locus can vary across tissues (11, 12). Local sequence preference, mismatches, bulges, and helix context contribute to this selectivity (13–17), but they do not specify whether a potential substrate forms, persists, localizes appropriately, or encounters the relevant ADAR state.

AdarEdit formalized intrinsic editability as graph learning over complete inverted-Alu substrates (18). Its compact graph attention network encodes nucleotide identity, local and global sequence, predicted secondary structure, pairing topology, geometry, thermodynamics, and motifs. A strict benchmark assigns entire Alu pairs globally to train, validation, or test, avoiding the severe leakage that would occur if sites from one duplex crossed partitions. The remaining errors could therefore reflect missing cellular context, underused supervision, pooling that dilutes the focal nucleotide, or an inadequate decomposition of substrate realization and local selectivity.

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Author contributions pending author confirmation.
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These explanations cannot be distinguished by adding one more annotation or a larger network. A biologically relevant variable can be redundant with sequence, measured at the wrong cellular scale, or too noisy to improve prediction. An architectural gain can likewise arise from overweighting substrates with many labeled sites rather than from a better account of editing. We therefore asked a broader question: within measurements available to the benchmark, where does incremental predictive information for endogenous A-to-I editing reside?

We preserved the published benchmark and examined biological context, supervision, substrate-wide and target-local representations, cross-tissue transfer, RNA foundation models, hierarchical formulations, and two forms of integration. Every discovery comparison used pair-disjoint development folds and whole-pair uncertainty; the canonical test remained sealed until one model was frozen. The resulting evidence separates four ideas. Whole-duplex sequence and structure encode intrinsic competence. Substrates have a measurable realization state, although current bulk context proxies estimate it poorly. A target and its helix partner retain local information that whole-duplex pooling loses. Finally, the duplex is not only a biological object but also the effective statistical unit for learning. ADARPro follows this hierarchy rather than increasing model capacity indiscriminately.

Results

A controlled framework separates biological scales. The benchmark comprises four individual GTEx tissue endpoints—artery tibial, brain/cerebellum, liver, and skeletal muscle—and Combined, a pooled endpoint rather than a fifth biological tissue. EditingIndex values at least 15% define positives and values below 1% define negatives; intermediate observations are excluded from the canonical binary task. The fixed global split contains 566 training, 139 validation, and 179 test pairs with zero overlap (Fig. 1).

Within the training and validation pairs, 5 outer folds generated out-of-fold (OOF) predictions for hypothesis testing and model selection. Feature fitting, checkpoint selection, and threshold choice were isolated from each outer fold. The internal reference was always trained and evaluated under the same folds as its candidate. Its lower absolute score reflects reduced training data and is not a replacement for published AdarEdit performance. The supplied-checkpoint canonical test remained the authoritative benchmark and was accessed only after the final model, thresholds, weights, code, and hashes had been frozen.

The campaign mapped hypotheses onto a biological hierarchy (Fig. 1A): cellular state can affect whether an inverted-Alu duplex is realized and engaged; the duplex determines intrinsic substrate competence; and local sequence and pairing influence which adenosine is edited. Outcomes were classified as selected, supported but not yet actionable, limited, not supported in the tested formulation, or deferred for lack of matched measurements (Fig. 2A). These categories distinguish biological relevance from incremental utility. Failure of bulk RBP features, for example, does not imply that RBPs do not regulate editing.

The substrate is a biological and statistical unit. The canonical row set samples a subset of the available observations, whereas the underlying training tables contain additional sites with valid high or low EditingIndex. Recovering these labels improves supervision, but uniform site sampling gives a substrate weight proportional to its number of observations. Sites within one duplex share sequence, structure, genomic context, and often editing behavior; they are not independent biological units.

A clean, existing same-fold ablation separated label recovery from pair weighting in muscle (Fig. 3A; SI Appendix). Adding all high/low observations under site-balanced sampling improved AUROC by +2.04 percentage points (95% pair interval +1.06 to +3.00) and F1 by +1.42 points (+0.21 to +2.57). Holding that expanded pool fixed and drawing a pair before class and site contributed a further +2.27 AUROC points (+1.25 to +3.36) and +2.85 F1 points (+1.68 to +4.01). Thus the gain is not reducible to either more labels or reweighting alone.

The combined pair-first formulation improved the same-fold reference in all five endpoints (Fig. 3B). All 15 AUROC, AUPRC, and F1 intervals excluded zero, with F1 gains from +0.69 to +4.23 per
result rather than a claim that the pair is a sin
pair-first sampling corrects the effective cont

The substrate also carries biological varia
separation between Alu arms, information d
them with a fixed separator, correlated neg
in every individual tissue (Spearman $\rho_s = -$
capacity models combining geometry with gra
edited fraction with $R^2 = 0.050$–0.277 and
alone in every tissue (Fig. 2C; SI Appendix).
measurable in these data.

Bulk context does not resolve measurable substra
actionable for site prediction proved harder
engagement gate added 0.0084 AUPRC but
included zero, and increased Brier score by
auxiliary likewise failed its promotion rule.
propensity, but it cannot reveal whether a
expected duplex, remained accessible, occu
encountered active ADAR.

The initial cellular-context hypothesis fo
combined replicated ENCODE eCLIP occu
predictions, literature-prioritized regulators,
by-abundance interactions, and randomize
Alu multimapping makes a missing eCLIP
predictors provided a complementary test
Neither class produced reproducible residual
switches were also nearly separated by tissu
available four tissue means from identifyin
mechanism. This finding constrains a bulk
does not contradict experimental evidence t
regulate editing (23–26).

Broad transcript, repeat-subfamily, diver
tations similarly failed to improve site-leve
Exact-tissue bulk H3K36me3 overlap produce
without stable AUROC, F1, or pair-engageme
H2A.Z resource supported a powered test.
therefore motivate quantitative, donor-matche
a claim that bulk peak overlap captures cotr
The coherent conclusion is not that substrate
that realization is measurable, whereas the a
proxies do not measure its cellular state p
strongest site classifier.

Target alignment resolves site-level selectivity. The c
a strong account of whole-duplex compet
embeddings can dilute the state of one targe
localized the missing information. Simple ta
ranking and calibration but not F1, and exp
node improved AUROC and AUPRC more
matched duplicate-global control. Both results
a specialized local encoder was built.

The selected local representation spans 65
focal adenosine and its dot-bracket-aligned
dilated convolution encodes nucleotide ide
partner indicators, and signed relative po
it reached AUROC/AUPRC/F1 of 0.9045
0.8885/0.8730/0.8208 for the pair-balanced v

The decisive control retained the same ar
shifted the local center by 96 nt. Performa
Aligned-minus-shifted differences were 0.192
0.1311 F1, with entirely positive pair-bootstra
capacity is therefore insufficient; the informat

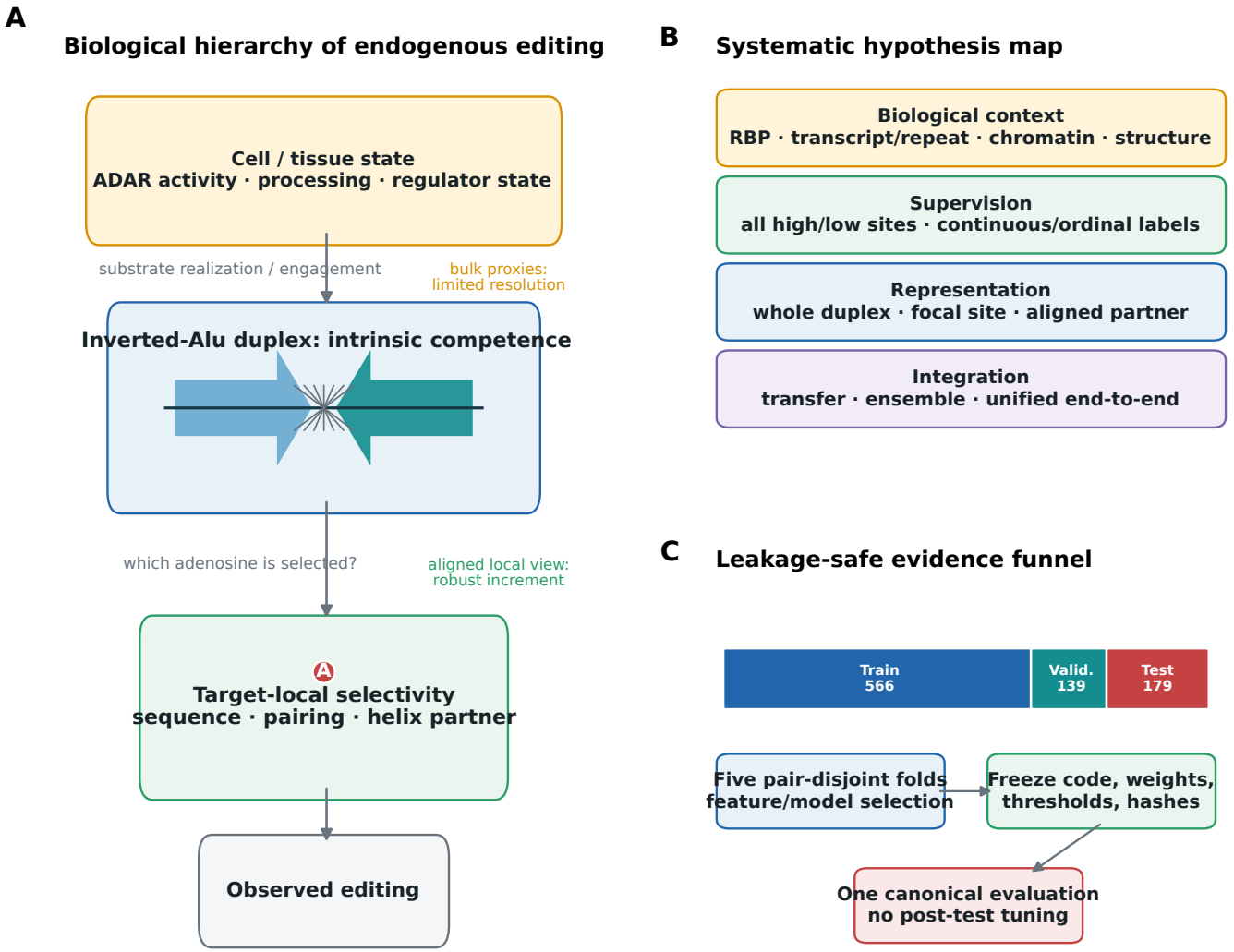


Fig. 1. A biological hierarchy and leakage-safe dissection of editing predictability. (A) Cell state modulates realization or engagement of an intrinsically competent inverted-Alu duplex; local sequence, pairing, and helix-partner context then contribute to selection of a particular adenosine. Labels indicate where the tested measurements yielded limited versus actionable information. (B) The campaign crossed biological context, supervision, representation, and integration hypotheses. (C) Whole pairs were partitioned globally. All discovery used pair-disjoint development folds; the final specification was frozen before one canonical-test evaluation. Combined was never assigned a fabricated tissue state.

its structural partner. Replication in liver and secondary natural-prevalence analysis supported the same direction.

Global substrate and target-local views are complementary. The aligned encoder did not make the whole-duplex graph redundant. The graph sees global sequence and structure, whereas the local component preserves target-centered detail. Their errors remained sufficiently distinct that a fixed mean of independently trained logits improved the pair-balanced graph by 0.0256/0.0308/0.0139 AUROC/AUPRC/F1 in muscle and 0.0182/0.0214/0.0179 in liver; all six intervals excluded zero (Fig. 3D).

ADARPro was consequently specified as two independently trained pair-first components: a whole-duplex graph model and a target/helix-partner-aligned model, combined by an unsearched 0.5/0.5 logit mean (Fig. 3E). This prediction-level integration tests complementarity without learning a flexible stacker on only hundreds of independent substrates.

Additional complexity provides limited returns. Intermediate EditingIndex values contain some graded information, but most useful thresholded content was already recovered by pair-balanced binary training. A binary/ordinal/continuous multitask head increased AUROC by 0.0081 (95% pair interval 0.0010–0.0151) while changing F1 by only 0.0008; its integration was adverse under natural prevalence. Shared four-tissue training likewis
macro changes of 0.0087 AUROC, 0.0114 AU
its prespecified promotion gates and worse
These limited results indicate that graded lab
ranking, but less than correcting supervision

Other plausible increases in capacity were
decoder trained all candidate adenosines joi
and F1 by -0.0599 relative to the pair-balan
futility boundary. Frozen RiNALMo target an
a larger model, did not add consistent valu
Larger or deeper graph networks likewise faile
outcomes do not imply that language-mode
intrinsically unsuitable; appropriately filtered
deferred because a benchmark-safe corpus co

Before test access, we preregistered one con
the surviving principles interact more effecti
combined a three-layer graph encoder, focal-
pairwise representation interactions, and a t
65,698 trainable parameters were fewer than
two ensemble components. On the 5,967-row,

A

Evidence map: what the benchmark can and cannot currently resolve

Selected · Supported · Limited · Not Supported · Deferred

Biological / learning scale	Tested formulation	Status	Interpretation
Substrate-aware learning	Pair-first expanded supervision	Selected	Corrects label use and pair weighting
Target selectivity	Aligned target + helix-partner view	Selected	Survives shifted-window control
Integration	Fixed equal-logit global + local	Selected	Complementary same-fold predictions
Substrate realization	Pair geometry / editing burden	Supported	State is measurable, not yet actionable
Supervision	Continuous + ordinal auxiliary labels	Limited	Ranking benefit; no useful F1 gain
Target selectivity	Focal node / contact geometry	Limited	Points toward aligned local view
Cross-tissue	Shared four-tissue training	Limited	Transfer below promotion gate
Substrate realization	Hierarchical engagement gate	Limited	Small ranking gain; unstable classification
Biological context	Bulk H3K36me3 overlap	Limited	Context-specific AUPRC only
Biological context	RBP binding × bulk expression	Not Supported	No reproducible residual gain
Biological context	Transcript / repeat annotation panel	Not Supported	No robust site-level increment
Representation	Frozen RNA-language embeddings	Not Supported	No consistent incremental utility
Representation	Substrate-wide all-A decoder	Not Supported	Stopped at prespecified futility gate
Capacity	Larger / deeper graph networks	Not Supported	Capacity alone did not generalize
Integration	Unified end-to-end architecture	Not Supported	Inferior to frozen ensemble
Cellular state	Matched donor/cell multimodality	Deferred	Required measurements unavailable
RNA state	Matched in-vivo structure / isoforms	Deferred	Required measurements unavailable

Status reflects prespecified same-fold comparisons and controls; deferred hypotheses were not tested.

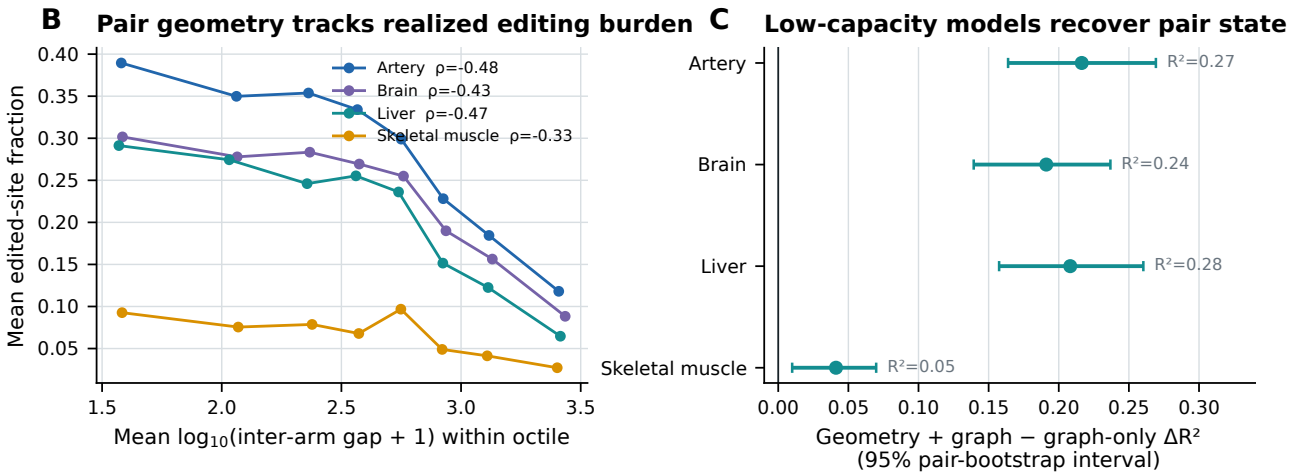


Fig. 2. A structured evidence map and positive evidence for substrate realization. (A) Hypotheses are grouped by biological or learning scale and by epistemic outcome. Status is based on prespecified same-fold comparisons and controls; "not supported" applies only to the tested formulation, and deferred measurements were not evaluated. Quantitative effects are reported only where estimands are commensurate. (B) Mean edited-site fraction across octiles of inter-arm gap in four tissues; ρ is calculated over individual development pairs. (C) Increment in pair-level R^2 from adding genomic geometry to graph-derived summaries. Bars are 95% whole-pair bootstrap intervals; labels give absolute R^2 for the combined model.

however, the unified model reached 0.9072/0.8922/0.8317 AUROC/AUPRC/F1 versus 0.9142/0.9040/0.8347 for the frozen ensemble. AUROC and AUPRC were worse in every fold; their pair-bootstrap intervals were wholly adverse, and Brier increased by 0.0205 (0.0135–0.0276). Prespecified repair and replication branches remained closed. At this effective sample size, separate optimization exploited complementary views more reliably than the tested joint representation.

The frozen hierarchy improves every canonical endpoint. After development selection, 10 component fits were trained on published training pairs. Published validation data alone selected checkpoints, epochs, thresholds, and one threshold per endpoint. The exact code revision, configurations, weights, and hashes were bound into a final manifest. A single authoriz test for one evaluation; no model or plain afterward.

ADARPro improved AUROC, AUPRC, and Macro gains over reproduced supplied-checkpo and 0.0253. F1 changed from 0.8671 to 0.88 brain, 0.8541 to 0.8812 in liver, 0.8139 to 0.85 to 0.8852 in Combined. Every context had a p interval for all three metrics, and Brier sc (Fig. 4C).

The gains were not purchased by indiscrim At frozen validation thresholds, ADARPro

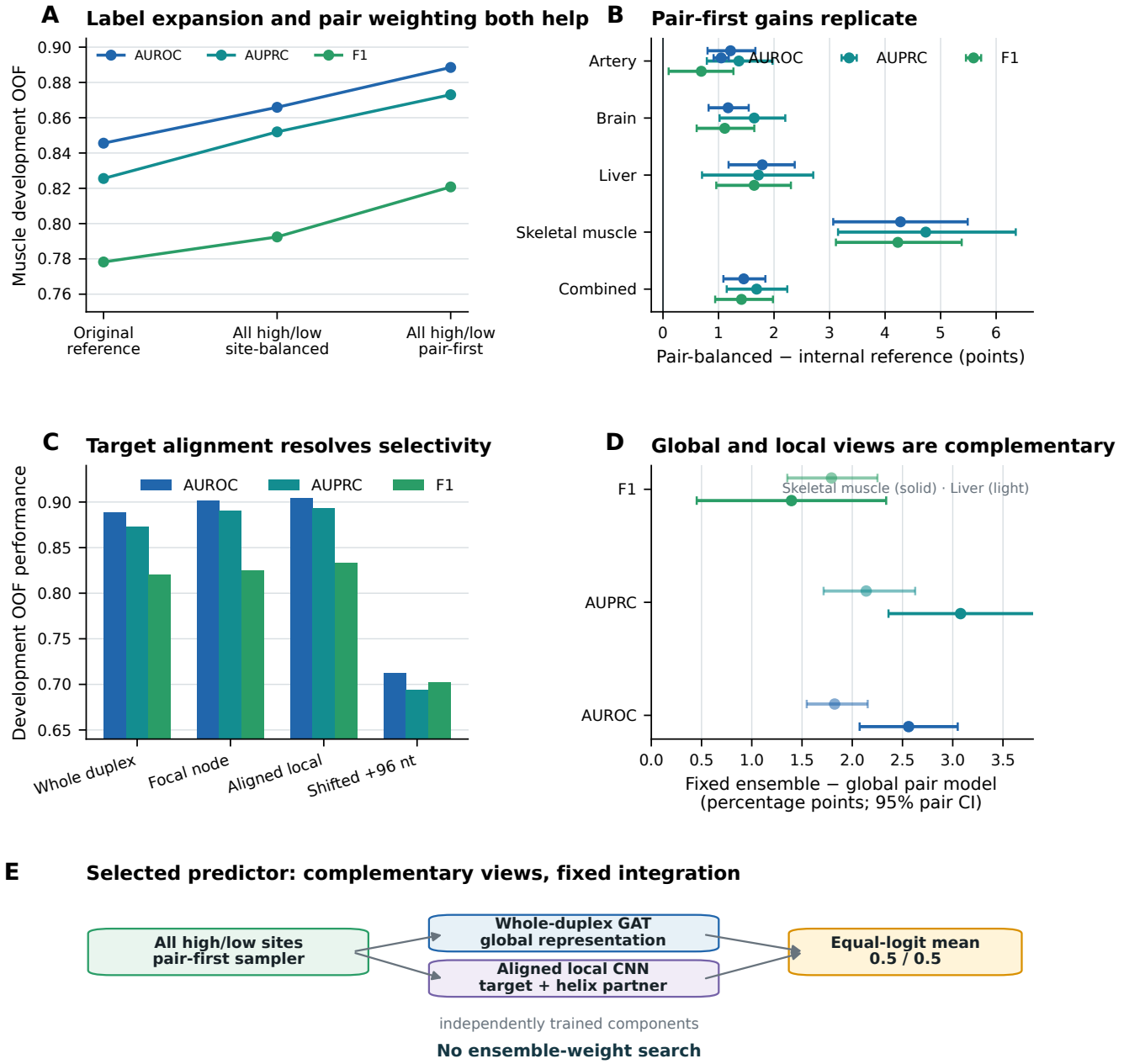


Fig. 3. The two actionable organizing principles and their integration. (A) Existing same-fold ablations separate the benefit of recovering all valid high/low labels from the additional benefit of pair-first weighting. (B) Pair-first expanded supervision improves the internal reference across all five endpoints; points and bars are pair-bootstrap medians and 95% intervals. (C) A progression from whole-duplex pooling through explicit focal state to the aligned target/partner representation; an identical encoder shifted by 96 nt fails. (D) The fixed equal-logit ensemble improves its whole-duplex component in muscle and liver. (E) Exact selected predictor. Components are trained independently; no ensemble weight, stacker, or post-test parameter is fitted. All panels show controlled development evidence, not canonical-test estimates.

false positives while damaging 1.7–5.9% of baseline true positives (Fig. 4D). At recall exactly matched to AdarEdit, the point estimates were 126, 119, 158, 68, and 148 fewer false positives across artery, brain, liver, skeletal muscle, and Combined; every clustered interval excluded zero (Fig. 4E). The one-use test thus validates the information hierarchy inferred entirely in development.

Discussion

This study asked what limits prediction of endogenous A-to-I editing after a strong sequence/structure model. The resulting picture is hierarchical. AdarEdit already captures substantial intrinsic duplex competence. The same benchmark nevertheless contains a separable substrate-realization dimension: inter-arm geometry covaries with editing burc
recover that state above graph-only summa
substrate, the fate of an adenosine remains
information that global pooling dilutes. Fin
a correlated statistical unit; learning improv
abundant site rows, control effective weight.

These levels should not be conflated. Pa
correction motivated by biological depende
in a substrate share one causal fate. Conv
positive biological observations even though
improve the best site classifier. Static geome
engagement also depends on transcription, spl

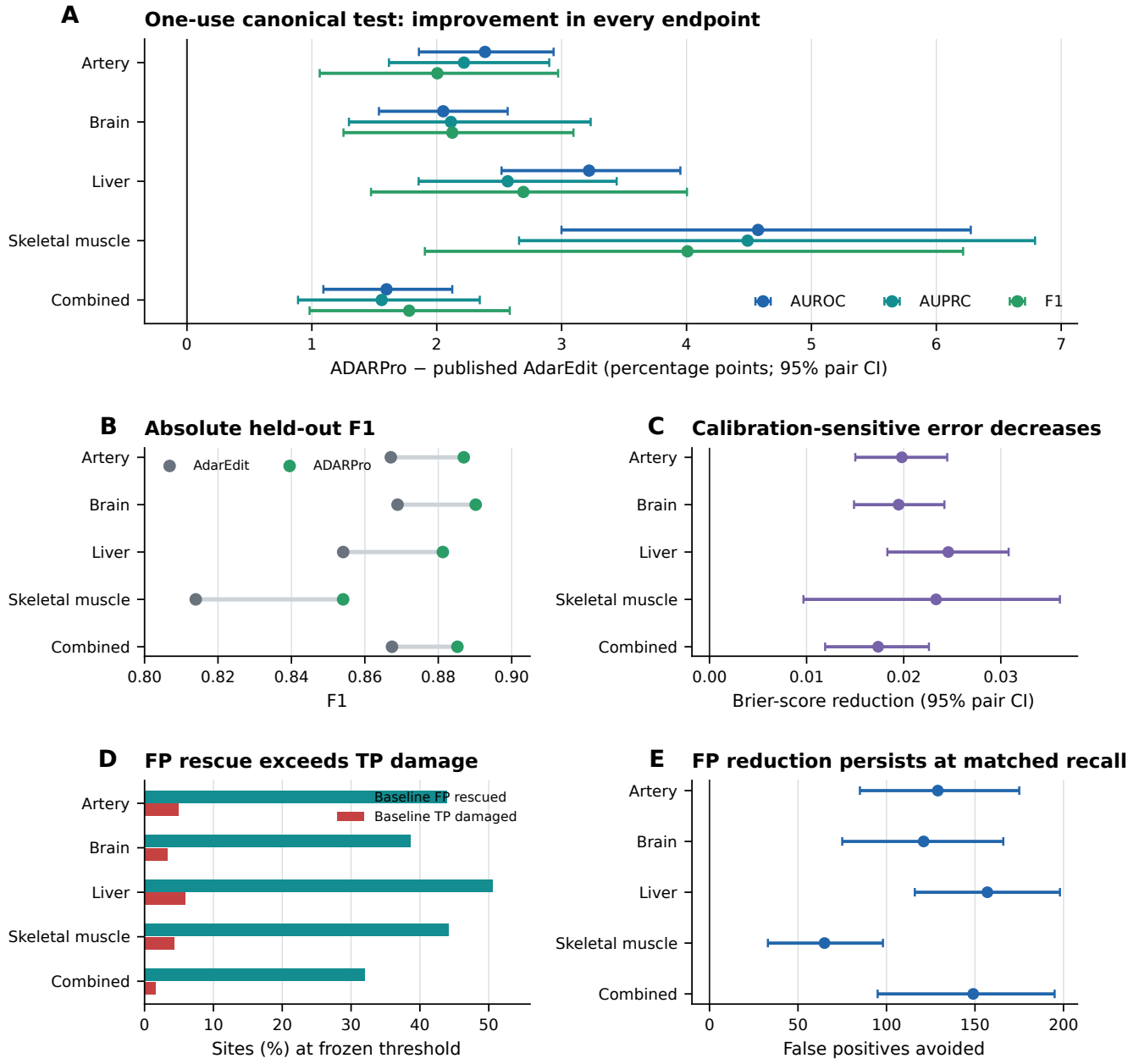


Fig. 4. Single-use canonical held-out validation. (A) ADARPro minus reproduced supplied-checkpoint AdarEdit for AUROC, AUPRC, and F1; bars are 95% whole-pair bootstrap intervals. (B) Absolute F1. (C) Reduction in Brier score, a calibration-sensitive error measure. (D) Baseline false positives corrected and baseline true positives damaged at frozen validation thresholds. (E) False positives avoided when ADARPro recall is matched exactly to AdarEdit. Combined is a pooled endpoint. All quantities were generated after the final manifest was frozen and the one-use authorization was consumed.

in-vivo conformation, ADAR isoform and activity, and regulator occupancy. Bulk tissue medians and binary peaks average over these states. They may therefore be biologically relevant yet predictively redundant or too mismatched at the site-by-sample level.

The negative and limited branches constrain rather than erase this biology. The tested RBP occupancy-by-expression representation did not add reproducible residual information, but cell-line CLIP and tissue-median abundance are not a direct assay of protein competition on a particular transcript. Bulk H3K36me3 overlap did not robustly improve classification, but this does not exclude cotranscriptional regulation. Cross-tissue sharing and continuous labels yielded small ranking effects, showing that transferable and graded structure exists without meeting the threshold for primary promotion. Likewise, frozen RNA-language embeddings and greater graph capacity did not substitute for correctly localized or biologically matched information.

The preregistered unified experiment quali
representation-level integration is not automat
level integration. With only hundreds of inde
a connected model may entangle views that
The result rejects the tested architecture, n
explains why the final predictor is deliberat
independently optimized, with no learned co

Several limitations define the scope of t
contains graph-buildable human inverted-Alu
not donor-resolved measurements or all editin
structure is a surrogate for the conformations
canonical task excludes intermediate observ
prevalence analyses are secondary. eCLIP
compromised by repeat mapping. Finally, th

but derives from the same benchmark construction; validation in an independent cohort remains necessary.

The canonical gain validates a scientific account rather than merely a leaderboard position. Better prediction followed from respecting substrate dependence and preserving target-centered context, not from indiscriminate feature or capacity growth. The major unresolved layer is realized substrate engagement. The next decisive experiment should measure editing, transcript abundance and processing, intron retention, ADAR isoform and activity, regulator state, chromatin, and in-vivo RNA structure in the same donors or cells and on the same substrates. Such data would directly test whether cellular realization can be separated from local adenosine selectivity.

Materials and Methods

Benchmark and labels. We used the AdarEdit v2 reproducibility release at upstream revision `5c8b57d70cdc2eb1c5e168431013275bcc9ed0c5`. The benchmark contains 884 graph-buildable inverted-Alu pairs. Human labels were edited for EditingIndex $\geq 15\%$, non-edited for EditingIndex $< 1\%$, and intermediate otherwise. The canonical task excludes intermediate observations. Pair identities and the fixed 566/139/179 train/validation/test assignment were verified before analysis. Combined was treated only as its published pooled endpoint.

Development protocol. All discovery used the original train plus validation pairs divided into 5 outer folds by pair ID. Learned preprocessing, feature selection, checkpoint choice, and threshold choice were fitted without the outer fold. Predictions were concatenated only after every row was scored by a model that had not trained on its pair. The test paths were protected by a stateful quarantine. Development OOF metrics estimate same-fold incremental effects; supplied-checkpoint test metrics remain the authoritative reference.

Models. The pair-balanced component retained the co
and trained from the expanded high/low site pool usin
pling. The local component encoded a 65-position targe
dilated convolutional layers and fused it with the graph
$\sigma[(\mathrm{logit}\, p_{\mathrm{pair}} + \mathrm{logit}\, p_{\mathrm{local}})/2]$; weights were fixed at 0
window control preserved architecture and optimization whi
definitions, architectures, ablations, and biological-context

Statistics and final evaluation. Metrics were AUROC
balanced accuracy, MCC, and Brier score. Thresholds m
training run. Uncertainty used 2,000 paired bootstrap rep
retained all their sites. For matched-recall analysis, the car
to equal the supplied-checkpoint baseline recall on final
pairs. Before final evaluation, code, inputs, weights, epo
values were frozen. Test authorization was consumed once
occurred. Natural-prevalence results use all eligible high
under unchanged pair assignments and are secondary to

Data, Materials, and Software Availability. Code, experime
figure scripts, retained derived artifacts, frozen checkpoin
//github.com/Scientific-Computing-Lab/ADARPro. Third-p
omitted from GitHub are listed with source URLs, roles, che
data manifest. The AdarEdit benchmark is available from h
AdarEdit. No new human participants were recruited ar
generated.

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